

$$\mathcal{K} = \mathbb{C} \oplus \mathcal{H} \oplus \hat{\mathcal{S}}(\mathcal{H} \otimes \mathcal{H}) \oplus \hat{\mathcal{S}}(\mathcal{H} \otimes \mathcal{H} \otimes \mathcal{H}) + \dots$$

$i=1, \dots, N$

• Phase space: $\{q_i\}, \{p_i\}$

$$[\hat{q}_i, \hat{p}_j] = i \delta_{ij} \hat{1}$$

$$\hat{q}_i |q\rangle = q_i |q\rangle$$

$$|q\rangle \equiv |q_1, \dots, q_N\rangle$$

$$\approx |q_1\rangle \otimes |q_2\rangle \otimes \dots \otimes |q_N\rangle$$

$$\hat{1} = \int d^N q |q\rangle \langle q|$$

$$= \int_{-\infty}^{\infty} dq_1 \int_{-\infty}^{\infty} dq_2 \dots \int_{-\infty}^{\infty} dq_N |q\rangle \langle q|$$

$$\mathcal{H} = \mathcal{H}_1^{\otimes N}$$

$$\langle q | q' \rangle = \langle q_1 \dots q_N | q'_1 \dots q'_N \rangle = \prod_{i=1}^N \delta(q_i - q'_i) = \delta^N(q - q')$$

$$|p\rangle = |p_1, \dots, p_N\rangle$$

$$\langle p | \psi \rangle = \psi(p) = \int d^N q e^{i \sum p_i q_i} \psi(q)$$

$$\langle p | \int d^N q |q\rangle \langle q | \psi \rangle$$

$$\int d^N q \langle p | q \rangle \langle q | \psi \rangle = \int d^N q \langle p | q \rangle \psi(q)$$

$$\langle p | q \rangle = e^{i \sum p_i q_i}$$

$$* \hat{1} = \int \frac{d^N p}{(2\pi)^N} |p\rangle \langle p|$$

$$\langle p | p' \rangle = \int d^N q e^{i p q} e^{-i q p'}$$

$$= \int d^N q e^{i q (p - p')} = (2\pi)^N \delta^N(p - p')$$

$$\langle p | p' \rangle = (2\pi)^N \delta(p - p')$$

$$* \int \frac{\langle q | p \rangle \langle p | q' \rangle d^N p}{(2\pi)^N} = \frac{1}{(2\pi)^N} \int d^N p e^{-i q p + i p q'}$$

$$\Rightarrow (2\pi)^N \delta^N(x) = \int d^N p e^{i p x}$$

or:

$$|q\rangle = \int \frac{d^N p}{(2\pi)^N} e^{-i \sum p_i q_i} |p\rangle$$

$$|p\rangle = \int d^N q e^{i \sum p_i q_i} |q\rangle$$

$$* [\hat{\phi}(x), \hat{\pi}(x')] = i \delta^d(x - x')$$

$$* \hat{\pi}(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

$$\hat{\phi}(x) |\phi(x)\rangle = \phi(x) |\phi(x)\rangle$$

• Replace cont. with lattice of points:

$$\vec{x}_{nml} = (nL, mL, lL)$$

$$\delta^3(\vec{x}_{nml} - \vec{x}'_{n'm'l'}) =$$

$$\delta(ax) = \frac{1}{|a|} \delta(x)$$

$$\delta^3(\vec{x}_{nml} - \vec{x}'_{n'm'l'}) = \frac{1}{L^3} \delta_{nn'} \delta_{mm'} \delta_{ll'}$$

$$* \text{Since } [\phi_{nml}, \hat{\pi}_{n'm'l'}] = i \delta^3(x_{nml} - x'_{n'm'l'})$$

$$[\hat{\phi}_{nml}, \hat{\pi}_{n'm'l'}] = \frac{i}{L^3} \delta_{nn'} \delta_{mm'} \delta_{ll'}$$

To get momenta from can. momenta:

$$[\hat{\phi}_{nme}, L^3 \hat{\pi}_{n'm'e'}] = i \delta_{nm} \delta_{m'm} \delta_{ee'}$$

$$\hat{p}_{n'm'e'} \equiv L^3 \hat{\pi}_{n'm'e'}$$

$$[\hat{\phi}_{nme}, \hat{p}_{n'm'e'}] = i \delta_{nm} \delta_{m'm} \delta_{ee'}$$

$$\hat{\phi}_{nme} |\phi\rangle = \phi_{nme} |\phi\rangle$$

$$\hat{1} = \left(\prod_{nme} \int_{-\infty}^{\infty} d\phi_{nme} \right) |\phi\rangle \langle \phi|$$

$$\langle \phi | \phi' \rangle = \prod_{nme} \delta(\phi_{nme} - \phi'_{n'm'e'})$$

$$\star \int \mathcal{D}\phi(x) \equiv \lim_{L \rightarrow 0} \left(\prod_{nme} \int_{-\infty}^{\infty} d\phi_{nme} \right)$$

$$\hat{1} = \left(\prod_{nme} \int d\phi_{nme} |\phi\rangle \langle \phi| \right)$$

$$\lim_{L \rightarrow 0} (\quad)$$

$$\hat{1} = \int \mathcal{D}\phi |\phi\rangle \langle \phi|$$

$$\langle \phi | \phi' \rangle = \delta_F(\phi - \phi')$$

$$\delta_F(\phi - \phi') \equiv \lim_{L \rightarrow 0} \prod_{nme} \delta(\phi_{nme} - \phi'_{nme})$$

Q: Why are the eigenstates of m . the conj. and not $\vec{p}$?

$$\hat{\Pi}(k_{nme}) |\pi\rangle = \pi_{nme} |\pi\rangle = \frac{1}{L^3} p_{nme} |\pi\rangle$$

$$\int D\pi(x) \equiv \lim_{L \rightarrow 0} \left(\prod_{nme} \int_{-\infty}^{\infty} \frac{d p_{nme}}{2\pi} \right)$$

$$\bullet p_{nme} = L^3 \pi_{nme}$$

$$\hat{1} = \int D\pi(x) |\pi\rangle \langle \pi|$$

$$\begin{aligned} \langle \pi | \pi' \rangle &= \delta_F(\pi(x) - \pi'(x)) \\ &\stackrel{\text{L.M.}}{=} \prod_{nme} (2\pi \delta(p_{nme} - p'_{nme})) \end{aligned}$$

$$\langle q | p \rangle = \prod_i \langle q_i | p_i \rangle = e^{i \sum_i q_i p_i}$$

$$\langle \phi | \pi \rangle = e^{i \sum_{nme} p_{nme} \phi_{nme}} = e^{i L^3 \sum_{nme} \pi_{nme} \phi_{nme}}$$

$$* \lim_{L \rightarrow 0} \left(L^3 \sum_{nme} \right) = \int d^3x$$

$$\langle \phi | \pi \rangle = e^{i \int d^3x \pi(x) \phi(x)}$$

* Similarly we have F.T.'s:

$$|\phi\rangle = \int D\pi(x) e^{-i \int d^3x \phi(x) \pi(x)} |\pi\rangle$$

$$|\pi\rangle = \int D\phi(x) e^{i \int d^3x \phi(x) \pi(x)} |\phi\rangle$$

$$\delta_F(\phi(x) - \phi'(x)) = \int D\pi(x) e^{i \int \pi(x) (\phi(x) - \phi'(x)) d^3x}$$

$$\text{Since } \langle \phi' | \phi \rangle = e^{i \int d^3x \pi(x) \phi(x)} = \int D\pi(x) e^{-i \int d^3x \phi \pi(x)} \langle \phi' | \pi \rangle$$

$$\delta_F(\phi'(x) - \phi(x)) = \int D\pi(x) e^{-i \int d^3x \phi(x) \pi(x)} e^{i \int d^3x \pi(x) \phi'(x)}$$

$$\delta_F(\phi'(x) - \phi(x)) = \int D\pi(x) e$$

$$f(\phi(x)) = \int D\phi' \delta_{\phi(x) - \phi'_x} f(\phi')$$

Functional derivative:

QM:

$$\psi(q) = \psi(q_1, \dots, q_N) = \langle q | \psi \rangle$$

$$\begin{aligned} \langle q | \hat{p}_j | \psi \rangle &= \int \frac{d^N p}{(2\pi)^N} \langle q | p \rangle \langle p | \hat{p}_j | \psi \rangle \\ &= \int \frac{d^N p}{(2\pi)^N} e^{i \sum_i q_i p_i} p_j \psi(p) \\ &= \int \frac{d^N p}{(2\pi)^N} \frac{1}{i} \frac{\partial}{\partial q_j} (e^{i \sum_i q_i p_i}) \psi(p) \\ &= \frac{1}{i} \frac{\partial}{\partial q_j} (\langle q | \psi \rangle) \\ &= -i \frac{\partial}{\partial q_j} \psi(q) \end{aligned}$$

$$\hat{p}_j(\psi)(q) = -i \frac{\partial \psi(q)}{\partial q_j}$$

$$\hat{p}_j \equiv -i \frac{\partial}{\partial q_j}$$

→ Since obs. are self-adjoint $p_j^\dagger = p_j$
 $\langle p | \hat{p}_j = \langle p | p_j$

$$\begin{aligned} * |q\rangle &\rightarrow |\phi\rangle \\ |p\rangle &\rightarrow |\pi\rangle \end{aligned}$$

→ wave functional: $\psi(\phi(x)) = \langle \phi | \psi \rangle$

$$\psi(\phi(x)) = \lim_{L \rightarrow \infty} \langle \{\phi_{nml}\} | \psi \rangle$$

$$\psi(\phi(x)) \in \mathbb{C} \otimes \mathbb{I}(M)$$

Space of sections over some S.t. mfd.

ex: $-\int d^3x (\phi(x))^2$

$\sim e$ on the lattice $\int d^3x \rightarrow L^3 \sum_{n \in \Lambda}$

$\phi \rightarrow \phi_{n \in \Lambda}$

$-\int d^3x \phi^2 = \lim_{L \rightarrow 0} e^{-L^3 \sum_{n \in \Lambda} \phi_{n \in \Lambda}^2}$

$\hat{\pi}_{n \in \Lambda} = \frac{1}{L^3} \hat{p}_{n \in \Lambda}$

Claim: $\langle \phi | \hat{p}_{n \in \Lambda} | \psi \rangle = -i \frac{\partial}{\partial \phi_{n \in \Lambda}} \langle \phi | \psi \rangle$

Proof:

$$\begin{aligned} \langle \phi | \hat{p}_{n \in \Lambda} | \psi \rangle &= \int D\pi(x) \langle \phi | \pi \rangle \langle \pi | \hat{p}_{n \in \Lambda} | \psi \rangle \\ &= \int D\pi(x) e^{i \int d^3x \phi(x) \pi(x)} \hat{p}_{n \in \Lambda} \langle \pi | \psi \rangle \\ &= \lim_{L \rightarrow 0} \left(\int \frac{d p_{n \in \Lambda}}{2\pi} e^{i \sum_{n \in \Lambda} \phi_{n \in \Lambda} p_{n \in \Lambda}} \psi(\pi) \right) \\ &= \lim_{L \rightarrow 0} \frac{1}{i} \frac{\partial}{\partial \phi_{n \in \Lambda}} \left(\int \frac{d p_{n \in \Lambda}}{2\pi} e^{i \sum \phi \pi} \psi(\pi) \right) \\ &= \frac{1}{i} \frac{\partial}{\partial \phi_{n \in \Lambda}} \langle \phi | \psi \rangle \end{aligned}$$

* In the limit $L \rightarrow 0$ care about π and not p :

$$\langle \phi | \hat{\pi}_{n \in \Lambda} | \psi \rangle = \frac{-i}{L^3} \frac{\partial \langle \phi | \psi \rangle}{\partial \phi_{n \in \Lambda}}$$

Functional derivative:

$$\frac{\delta}{\delta \phi(x)} \equiv \lim_{L \rightarrow 0} \frac{1}{L^3} \frac{\partial}{\partial \phi_{n \in \Lambda}}$$

$$\langle \phi | \hat{\pi}(x) | \psi \rangle = -i \frac{\delta}{\delta \phi(x)} \langle \phi | \psi \rangle$$

ex: Consider

$$F(\phi) = \int d^3x J(x) \phi(x)$$

$$F = L^3 \sum_{nml} J(x_{nml}) \phi_{nml}$$

$$\frac{\delta F}{\delta \phi(x)} = \lim_{L \rightarrow 0} \left(\frac{1}{L^3} \frac{\partial}{\partial \phi_{nml}} L^3 \sum_{n'm'l'} J_{n'm'l'} \phi_{n'm'l'} \right)$$

$$= \sum_{\substack{n'm'l' \\ j_{nml}}} \delta_{nn'} \delta_{mm'} \delta_{ll'} J_{n'm'l'}$$

$$= \lim_{L \rightarrow 0} J_{nml}$$

$$= J(x)$$

$$\boxed{\frac{\delta F}{\delta \phi(x)} = J(x)}$$

Can invert this: $\lim_{L \rightarrow 0} \frac{1}{L^3} \frac{\partial F}{\partial \phi_{nml}} = \lim_{L \rightarrow 0} J_{nml}$

$$F(\phi) = \text{const.} + L^3 \sum_{nml} \phi_{nml} J_{nml}$$

$$\lim_{L \rightarrow 0} () \Rightarrow F(\phi) = \text{const.} + \int d^3x \phi(x) J(x)$$

$$\bullet F[\phi(x)] = \int d^3x \int d^3x' \phi(x) K(x, x') \phi(x')$$

$$F[\phi_{nml}] = L^3 \sum_{nml} \sum_{n'm'l'} \phi_{nml} K(x_{nml}, x'_{n'm'l'}) \phi_{n'm'l'}$$

$$\frac{\delta F}{\delta \phi(x)} = \lim_{L \rightarrow 0} \frac{1}{L^3} \frac{\partial}{\partial \phi_{nml}} \left(L^6 \sum_{n'} \sum_{n''} \phi_{n'm'l'} K(x_{nml}, x'_{n'm'l'}) \phi_{n'm'l''} \right)$$

$$\frac{\delta F}{\delta \phi} = \lim_{L \rightarrow 0} L^3 \sum_{n'n''} \delta_{nn'} \delta_{mm'} \delta_{ll'} K(x_n, x'_{n''}) \phi_{n''m'l''}$$

$$+ \phi_{n'm'l'} K(x_{n'm'l'}, x'_{n''m'l''}) \delta_{nn''} \delta_{mm''} \delta_{ll''}$$

$$= \lim_{L \rightarrow 0} L^3 \left(\sum_{n''} K(x_{nml}, x'_{n''m'l''}) \phi_{n''m'l''} \right.$$

$$\left. + \sum_{n'} \phi_{n'm'l'} K(x_{nml}, x'_{n'm'l'}) \right)$$

$$= \int d^3x' K(x, x') \phi(x') + \int d^3x' K(x', x) \phi(x')$$

$$\frac{\delta F}{\delta \phi(x)} = \int d^3x' (K(x, x') \phi(x') + \phi(x') K(x', x))$$

* Define: $\delta F = \epsilon \rightarrow 0 \left(\frac{F(\phi(x) + \epsilon \delta \phi(x)) - F(\phi(x))}{\epsilon} \right)$

$$Z(\phi(x)) = \int \mathcal{D}\pi e^{-i \int d^4x \phi(x) \pi(x)}$$

$$F(\phi + \epsilon \delta) \approx F(\phi) + \epsilon \int d^4x \delta \phi(x) \frac{\delta F}{\delta \phi}$$

$$\delta F = \int d^3x \left(\frac{\delta F}{\delta \phi(x)} \right) \delta \phi(x)$$

ex: $F[\phi] = \int d^3x J(x) \phi(x)$

$$\delta F = \lim_{\epsilon \rightarrow 0} \left(\frac{\int d^3x J(x) [\phi + \epsilon \delta \phi] - \int d^3x J(x) \phi(x)}{\epsilon} \right)$$

$$= \int d^3x J(x) \delta \phi$$

$$\frac{\delta F}{\delta \phi} = J(x)$$

ex: $\delta F = \lim_{\epsilon \rightarrow 0} \left(\frac{F(\phi + \epsilon \delta \phi) - F(\phi)}{\epsilon} \right)$

$$\rightarrow F[\phi] = \int d^3x' d^3x \phi(x) K(x, x') \phi(x')$$

$$\delta F = \int d^3x d^3x' (\phi + \epsilon \delta \phi(x)) K(x, x') (\phi(x') + \epsilon \delta \phi(x'))$$

$$- \int d^3x d^3x' \phi(x) K(x, x') \phi(x') / \epsilon$$

$$= \int d^3x d^3x' (\delta \phi(x) K(x, x') \phi(x') + \phi(x) K(x, x') \delta \phi(x'))$$

$$= \int d^3x d^3x' (K(x, x') \phi(x') + \phi(x') K(x', x)) \delta \phi(x)$$

$$\frac{\delta F}{\delta \phi(x)} = \int d^3x' (K(x, x') \phi(x') + \phi(x') K(x', x))$$

$$\phi(y) = \int d^3x \phi(x) \delta^3(x-y)$$

$$\delta \phi(y) = \int d^3x \delta^3(x-y) \delta \phi(x)$$

$$\frac{\delta \phi(y)}{\delta \phi(x)} = \delta^3(x-y)$$

$$\begin{aligned} * \nabla_y^2 \phi(y) &= \int d^3x \delta^3(x-y) \nabla_x^2 \phi(x) \\ &= \int d^3x \nabla_x^2 \delta^3(x-y) \phi(x) \end{aligned}$$

$$\delta(\nabla_y^2 \phi(y)) = \int d^3x \nabla_x^2 \delta^3(x-y) \delta \phi(x)$$

$$\frac{\delta(\nabla_y^2 \phi(y))}{\delta \phi(x)} = \nabla_x^2 \delta^3(x-y)$$

$$\text{or} \quad \frac{\delta}{\delta \phi(y)} \circ \nabla_y^2 = \nabla_x^2 \circ \frac{\delta}{\delta \phi(x)}$$

Free-scalar field:

$$\hat{\phi}(x, t) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \left(\hat{a}_k e^{ikx - i\omega_k t} + \hat{a}_k^\dagger e^{-ikx + i\omega_k t} \right)$$

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = (2\pi)^3 2\omega_k \delta^3(k - k')$$

$$\omega_k = \sqrt{k^2 + m^2}$$

$$\hat{\mathcal{H}}(x, t) = \frac{\partial \hat{\phi}(x, t)}{\partial t} \quad \left| \quad \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right.$$

$$\begin{aligned} \hat{\mathcal{H}}(x, t) &= \int \frac{d^3k}{(2\pi)^3} \left(\frac{(-i\omega_k) \hat{a}_k e^{-ikx - i\omega_k t}}{2\omega_k} + \hat{a}_k^\dagger \left(\frac{i\omega_k}{2\omega_k} \right) e^{-ikx + i\omega_k t} \right) \\ &= \int \frac{d^3k}{2\omega_k} \left(-i\omega_k e^{+ikx - i\omega_k t} \hat{a}_k + i\omega_k \hat{a}_k^\dagger e^{-ikx + i\omega_k t} \right) \end{aligned}$$

Sch. picture:

$$\hat{\phi}_s(\vec{x}) = \hat{\phi}(x, 0)$$

$$\hat{\pi}_s(\vec{x}) = \hat{\pi}(x, 0)$$

$$\sqrt{-\nabla^2 + m^2} \hat{\phi}_s(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \sqrt{m^2 + k^2} e^{ikx} a_k + e^{-ikx} \tilde{a}_k$$
$$= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2} \left(\hat{a}_k e^{ikx} + \hat{a}_k^\dagger e^{-ikx} \right)$$

$$i\hat{\pi}(\vec{x}) = \int d^3k \frac{1}{2} a_k e^{ikx} - \frac{1}{2} \tilde{a}_k e^{-ikx}$$

$$\sqrt{-\nabla^2 + m^2} \phi_s(x) + i\hat{\pi}(x) = \int d^3k \hat{a}_k e^{ikx}$$

$$\hat{a}_k = \int d^3x e^{-ikx} \left(\sqrt{-\nabla^2 + m^2} \phi(x) + i\hat{\pi}(x) \right)$$

$$\hat{a}_k = \int d^3x e^{-ikx} \left(\sqrt{-\nabla^2 + m^2} \phi(x) + i\hat{\pi}(x) \right)$$

Now,

$$\langle \phi | \hat{\phi}(\vec{x}) | \psi \rangle = \phi(x) \langle \phi | \psi \rangle$$

$$\langle \phi | \hat{\pi}_\alpha | \psi \rangle = -i \frac{\delta}{\delta \phi(x)} \langle \phi | \psi \rangle$$

$$\hat{a}_k |0\rangle = 0 \quad \forall k$$

$$\int d^3x e^{-ikx} \left(\sqrt{-\nabla^2 + m^2} \phi(x) + i\hat{\pi}(x) \right) |0\rangle = 0$$

$$\left(\sqrt{-\nabla^2 + m^2} \phi(x) + i\hat{\pi}(x) \right) |0\rangle = 0$$

$$\langle \phi | \left(\sqrt{-\nabla^2 + m^2} \hat{\phi}(x) + i\hat{\pi}(x) \right) |0\rangle = 0$$

$$\langle \phi | \sqrt{-\nabla^2 + m^2} \hat{\phi} |0\rangle + \frac{\delta}{\delta \phi(x)} \langle \phi |0\rangle = 0$$

$$\boxed{\left(\sqrt{-\nabla^2 + m^2} \phi(x) + \frac{\delta}{\delta \phi(x)} \right) \langle \phi |0\rangle = 0}$$

$$\phi(x) = \int d^3y$$

$$\sqrt{-\nabla^2 + m^2} \phi(x) = \int d^3y D(x, y) \phi(y)$$

$$\phi(x) = \int d^3y \delta^3(x-y) \phi(y)$$

$$\Rightarrow \sqrt{-\nabla^2 + m^2}$$

$$(\nabla^2 + m^2) \phi = 0$$

$$\nabla^2 \phi = -m^2 \phi$$

$$\phi(x) = \int d^3y \delta^3(x-y) \phi(y)$$

$$\nabla^2 \phi = \int d^3y \nabla^2 \delta^3(x-y) \phi(y) \equiv \int d^3y D(x, y) \phi(y)$$

Aside $Lu = f$ in $\Omega \subset \mathbb{R}^3$

$$LG(x, \xi) = \delta(x - \xi)$$

$$\int LG(x, \xi) f(\xi) dV = \int \delta(x - \xi) f(\xi) dV$$

$$= L\left(\int_V G(x, \xi) f(\xi) dV\right) = f(x)$$

$$u = \int D(x, x') f(x') d^3x'$$

$$D(x, x') \equiv \sqrt{-\nabla_x^2 + m^2} \delta^3(x - x') = \sqrt{-\nabla_x^2 + m^2} \int \frac{d^3k}{(2\pi)^3} e^{ik(x-x')} \\ = \int \frac{d^3k}{(2\pi)^3} \omega_k e^{ik(x-x')}$$

$$\int d^3x' D(x, x') \phi(x') = \sqrt{-\nabla_x^2 + m^2} \phi(x)$$

$$D(x, x') = D(x', x)$$

$$\int \dots \delta \dots$$

$$\left(\int d^3x' D(x, x') \phi(x') + \frac{1}{L^3} \frac{\partial}{\partial \phi(x)} \right) \langle \phi | 0 \rangle$$

$$\left(L^3 \sum_{n \neq n'} D(x_{nne}, x_{n'ne}) \phi_{nne} + \frac{1}{L^3} \frac{\partial}{\partial \phi_{nne}} \right) \langle \phi | 0 \rangle = 0$$

$$\int L^6 \sum_n D_{nn'} \phi_n \delta \phi_n = - \int \partial (\langle \phi | 0 \rangle) / \langle \phi | 0 \rangle$$

$$- \frac{1}{2} L^6 \sum_n \sum_{n'} \phi_n D_{nn'} \phi_{n'} = \ln (\langle \phi | 0 \rangle)$$

$$\langle \phi | 0 \rangle = A e^{-\frac{1}{2} \int d^3x d^3x' \phi(x) D(x, x') \phi(x')}$$

Determines the Vacuum

$$\hat{1} = \int D\phi |\phi\rangle \langle \phi| = \lim_{L \rightarrow 0} \left(\prod_{nne} \int_{-\infty}^{\infty} d\phi_{nne} |\phi\rangle \langle \phi| \right)$$

$$\langle 0 | 0 \rangle = \int D\phi \langle 0 | \phi \rangle \langle \phi | 0 \rangle$$

$$\langle 0 | 0 \rangle = |A|^2 \int D\phi e^{-\int d^3x d^3x' \phi(x) D(x, x') \phi(x')}$$

* Transition using Gaussian integrals

$$\langle 0 | 0 \rangle = \lim_{L \rightarrow 0} \left(\prod_{nne} \int_{-\infty}^{\infty} d\phi_{nne} e^{-L^6 \sum_n \sum_{n'} \phi_{nne} D_{nn'} \phi_{n'ne}} \right)$$

$A = A^T$ is a symm matrix, i.e. $D(x, x') = D(x', x)$

$$I = \int d^n \phi e^{-\phi^T A \phi}$$

$$\bullet A = O A_{diag} O^T$$

$$O^T O = I \quad | \det I | = 1$$

$$A^T A^{-1} \quad A^T A^{-1} = I$$

$$\phi^T A \phi = \phi^T O A_d O^T \phi$$

$$\left| \begin{aligned} \tilde{\phi} &\equiv O^T \phi \quad \Rightarrow \quad \tilde{\phi}^T = \phi^T O \\ \hookrightarrow &= \tilde{\phi}^T A_d \phi \end{aligned} \right.$$

$$\int d^N \phi \rightarrow d\phi^2 = (O \tilde{\phi})^2 \\ = d(O_{\alpha}^{\pm}, \tilde{\phi}^{\alpha})$$

$$d\phi^2 = O_{\alpha}^{\pm} d\tilde{\phi}^{\alpha}$$

$\Rightarrow$

$$d\phi^2 \wedge d\phi^2 \wedge \dots \wedge d\phi^N = O_{\alpha_1}^{\pm} O_{\alpha_2}^{\pm} \dots O_{\alpha_N}^{\pm} d\tilde{\phi}^{\alpha_1} \wedge \dots \wedge d\tilde{\phi}^{\alpha_N}$$

$$= O_{\alpha_1}^{\pm} \delta_1^{\alpha_1} \dots \delta_N^{\alpha_N} d\tilde{\phi}^{\alpha_1} \wedge \dots \wedge d\tilde{\phi}^{\alpha_N}$$

$$= \sum_{\alpha_1 \dots \alpha_p} \epsilon^{\beta_1 \dots \beta_p} = -p! \sum_{\delta \alpha_1 \dots \alpha_p}$$

$$= -p! \left(\delta_{\alpha_1}^{\beta_1} \dots \delta_{\alpha_p}^{\beta_p} \right)$$

$$= -$$

$$O_{\alpha_1}^{\pm} \dots O_{\alpha_N}^{\pm} \left(\underbrace{-\epsilon_{1 \dots N}}_{-1} \epsilon^{\alpha_1 \dots \alpha_N} \right) d\tilde{\phi}^{\alpha_1} \wedge \dots \wedge d\tilde{\phi}^{\alpha_N}$$

$$\sum_{\alpha_1 \dots \alpha_N} O_{\alpha_1}^{\pm} \dots O_{\alpha_N}^{\pm} = \det(O)$$

$$d^N \phi = \det(O) d^N \tilde{\phi}$$

$$\det(O) = 1 \quad \Rightarrow \quad d^N \phi = d^N \tilde{\phi}$$

Thus;

$$\int d^N \tilde{\phi} e^{-\tilde{\phi}^T A_d \tilde{\phi}} = \int d\phi^1 \dots d\phi^N e^{-\alpha_1 (\tilde{\phi}^1)^2 - \alpha_2 (\tilde{\phi}^2)^2 - \alpha_3 (\tilde{\phi}^3)^2 - \dots}$$

$$A = \begin{pmatrix} \alpha_1 & \alpha_2 & 0 \\ 0 & \dots & \alpha_N \end{pmatrix} \quad \sqrt{\frac{\pi}{\alpha_i}}$$

$$\begin{aligned} \int d\vec{\phi} e^{-\alpha_i (\vec{\phi})^2} &= \prod_{i=1}^N \left(\int d\phi e^{-\alpha_i \phi_i^2} \right) \\ &= \prod_{i=1}^N \left(\sqrt{\frac{\pi}{\alpha_i}} \right) = \prod_{i=1}^N \frac{1}{\sqrt{\frac{\alpha_i}{\pi}}} \\ &= \frac{1}{\sqrt{\prod_{i=1}^N \left(\frac{\alpha_i}{\pi} \right)}} = \frac{1}{\sqrt{\text{Det}\left(\frac{A}{\pi}\right)}} \end{aligned}$$

$$\int d^N \phi e^{-\phi^T A \phi} = \left(\text{Det}\left(\frac{A}{\pi}\right) \right)^{-1/2}$$

* Normalize the vacuum state $\langle 0|0 \rangle \stackrel{!}{=} 1$

$$|A|^2 \int D\phi e^{-\left(\int d^3x d^3x' \phi(x) D(x, x') \phi(x') \right)} = 1$$

$$|A|^2 \lim_{L \rightarrow \infty} \prod_{nm} \int d\phi_{nm} e^{-L^6 \sum_n \sum_{n'} \phi_n D_{nn'} \phi_{n'}} = 1$$

$$|A|^2 \lim_{L \rightarrow \infty} \left(\text{Det} \left(\frac{L^6 D(x_{nm}, x_{n'm'})}{\pi} \right)^{-1/2} \right) = 1$$

$$|A| = \lim_{L \rightarrow \infty} \text{Det} \left(\frac{L^6 D(x_{nm}, x_{n'm'})}{\pi} \right)^{1/6}$$

* I'll define

$$\langle \phi | k \rangle = \langle \phi | \hat{a}_k^\dagger | 0 \rangle$$

$$= \int d^3k e^{-ikx} \left(\sqrt{-\nabla^2 + m^2} \phi(x) - \frac{\delta}{\delta \phi} \right) \langle \phi | 0 \rangle = 0$$

$$\langle \phi | 0 \rangle = A e^{-\frac{1}{2} \left(\int d^3y d^3y' \phi(y) D(y, y') \phi(y') \right)}$$

$$\begin{aligned} \frac{\delta \langle \phi | 0 \rangle}{\delta \phi(x)} &= A \left(-\frac{1}{2} \int d^3x' D(x, x') \phi(x') + -\frac{1}{2} \int d^3y \phi(y) D(x, y) \right) e^{-\frac{1}{2} \int \phi D \phi} \\ &= A \left(-\int d^3x' D(x, x') \phi(x') \right) e^{-\frac{1}{2} \int \phi D \phi} \end{aligned}$$

$$= - \int d^3x' D(x, x') \phi(x') \langle \phi | 0 \rangle$$

$$= -\sqrt{-\partial^2 + m^2} \phi(x) \langle \phi | 0 \rangle$$

$$\begin{aligned} \langle \phi | \psi \rangle &= \int d^3k e^{-ikx} 2\sqrt{-\partial^2 + m^2} \phi(x) \langle \phi | 0 \rangle \\ &= \int d^3k e^{-ikx} 2\omega_k \phi(x) \langle \phi | 0 \rangle \\ &= \phi(x) \int d^3k e^{-ikx} 2\omega_k \langle \phi | 0 \rangle \\ &= 2(2\pi)^3 D(x, x) \langle \phi | 0 \rangle \end{aligned}$$

$$|\psi\rangle = \int d^3x_1 \dots d^3x_N \psi(\phi_1, \dots, \phi_N) a_{\vec{x}_1}^\dagger \dots a_{\vec{x}_N}^\dagger |0\rangle$$

$$\langle \phi | \psi \rangle = \int d^3x_1 \dots d^3x_N \psi(\phi_1, \dots, \phi_N) \langle \phi | a_{\vec{x}_1}^\dagger \dots a_{\vec{x}_N}^\dagger | 0 \rangle$$

• a_1 :

$$\langle \phi | \psi \rangle = \int d^3x_1 \psi[\phi] \langle \phi | \hat{a}_n | 0 \rangle$$

$$\psi[\phi] = \int d^3x_1 \psi[\phi]$$

* Formal der. of Path int in QM:

2D phase space $T^*\mathcal{M}$

$(q_1, \dots, q_N, p_1, \dots, p_N)$

- $\{q_i, p_j\} = \delta_{ij}$
- $[q_i, p_j] = i\hbar \delta_{ij}$
- $[q_i, q_j] = 0$
- $[p_i, p_j] = 0$

$$* \hat{q}_i |q\rangle = q_i |q\rangle$$

$$|\psi\rangle = \sum_i \lambda_i |q_i\rangle$$

$$\mathbb{I} |\psi\rangle = \int dq \langle q | \psi \rangle |q\rangle$$

$$|\psi\rangle = \int dq \psi(q) |q\rangle$$

• $\hat{q}_i$ is "diagonalized":

$$\hat{q}_i |q\rangle = q_i |q\rangle$$

$$\hat{q}_i |q\rangle = q_i \int d\tilde{q} (\tilde{q} | \langle \tilde{q} | q \rangle) \\ = \left(\int d\tilde{q} q_i |\tilde{q}\rangle \langle \tilde{q}| \right) |q\rangle$$

$$\leadsto \hat{q}_i = \int d\tilde{q} q_i |q\rangle \langle q|$$

$$\hat{p}_i |p\rangle = p_i |p\rangle$$

$$\hat{p}_i = \int \frac{d^D p}{(2\pi\hbar)^D} p_i |p\rangle \langle p|$$

$$|q\rangle = \int \frac{d^D p}{(2\pi\hbar)^D} |p\rangle e^{-i \sum_i q_i p_i / \hbar}$$

$$|p\rangle = \int d^D q |q\rangle e^{i \sum_i q_i p_i / \hbar}$$

$$\langle q | q' \rangle = \delta^D(q - q') = \prod_i \delta(q_i - q'_i)$$

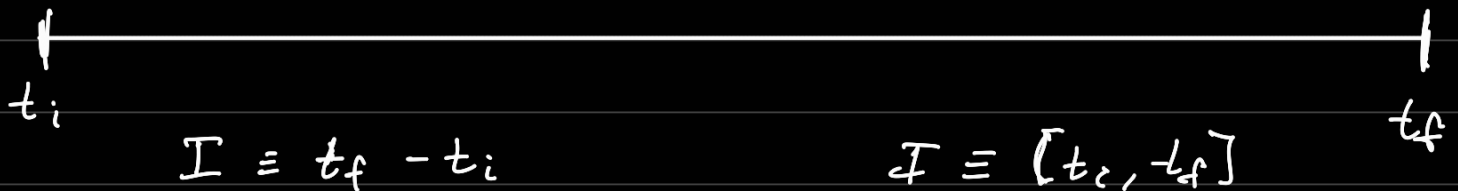
$$\langle p | p' \rangle = (2\pi)^D \delta^D(p - p') = \prod_i 2\pi \delta(p_i - p'_i)$$

Time evolution: $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$

$$\hat{U}(t_f, t_i) = e^{-i\hat{H}(t_f - t_i)/\hbar}$$

$$|\psi(t_f)\rangle = \hat{U}(t_f, t_i) |\psi(t_i)\rangle$$

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H} |\psi(t)\rangle$$



t_i t_f

$I \equiv t_f - t_i$ $T \equiv [t_i, t_f]$

Each step: $\Delta t \equiv \frac{I}{N}$

0th step: t_i

1st step: $t_1 = t_i + \Delta t$

2nd step: $t_2 = t_i + 2 \Delta t$

n^{th} step $t_n = t_i + n \Delta t$

$$U(t_f, t_i) = e^{-i\hat{H}(t_f - t_i)/\hbar} = e^{-i\hat{H}N\Delta t/\hbar} \\ = (e^{-i\hat{H}\Delta t/\hbar})^N$$

* Let's see how states can evolve from

$$\langle q_N | \hat{U}(t_f, t_i) | q_0 \rangle$$

$$\langle q_N | \hat{U}(t_f, t_i) \int \frac{d^D p}{(2\pi)^D} | p_0 \rangle \langle p_0 | q_0 \rangle \\ = \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \langle q_N | \hat{U}(t_f, t_i) | p_0 \rangle$$

$$t_N = t_i + N \left(\frac{t_f - t_i}{N} \right) = t_f$$

$$= \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \langle q_N | \hat{U}(t_N, t_i) | p_0 \rangle$$

$$= \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \langle q_N | \hat{U}(\Delta t) \dots \hat{U}(\Delta t) | p_0 \rangle$$

$$\hat{1} = \left(\int \frac{d^D p_n}{(2\pi)^D} | p_n \rangle \langle p_n | \right) \left(\int d^D q_n | q_n \rangle \langle q_n | \right)$$

$$= \int \frac{d^D p_n}{(2\pi)^D} \int d^D q_n e^{-i p_n q_n} | p_n \rangle \langle q_n |$$

$$\int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \left(\int \frac{d^D p_{N-1}}{(2\pi)^D} \int d^D q_{N-1} \langle q_N | \hat{U}(\Delta t) | p_{N-1} \rangle \langle q_{N-1} | \hat{U}(\Delta t) | p_{N-1} \rangle \right) \dots \langle q_1 | \hat{U}(\Delta t) | p_0 \rangle$$

$$\langle q_N | \hat{U}(t_N, t_i) | q_0 \rangle = \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \prod_{n=0}^{N-1} \int \frac{d^D p_i}{(2\pi)^D} d^D q_i \langle q_{n+1} | \hat{U}(\Delta t) | p_n \rangle$$

$$\int d\Omega \equiv \prod_{n=1}^{N-1} \int \frac{d^D p_n d^D q_n}{(2\pi)^D}$$

$$\left(e^{-i \sum_n q_n p_n} \right)$$

$$\langle q_N | \hat{U}(t_N, t_i) | q_0 \rangle = \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \left[\int d\Omega \left(\prod_{n=0}^{N-1} \langle q_{n+1} | \hat{U}(\Delta t) | p_n \rangle \right) \times e^{-i \sum_{n=1}^{N-1} q_n p_n} \right]$$

* look at:

$$\begin{aligned} \langle q_{n+1} | \hat{U}(\Delta t) | p_n \rangle &= \langle q_{n+1} | e^{-i \Delta t \hat{H} / \hbar} | p_n \rangle \\ &= \langle q_{n+1} | \left(\hat{1} - \frac{i \Delta t \hat{H}}{\hbar} + \mathcal{O}\left(\frac{1}{N^2}\right) \right) | p_n \rangle \end{aligned}$$

• Harmonic oscillator:

$$\hat{H} = \sum_i \frac{\hat{p}_i^2}{2m} + V(\hat{q}_i) = \hat{H}(q_i, \hat{p}_i)$$

$$\begin{aligned} \langle q_{n+1} | \hat{H} | p_n \rangle &= \left(\sum_i \frac{p_i^2}{2m} + V(q_i) \right) \langle q_{n+1} | p_n \rangle \\ &= \langle q_{n+1} | p_n \rangle \left(1 - \frac{i \Delta t}{\hbar} \left(\sum_i \frac{p_i^2}{2m} + V(q_i) \right) + \mathcal{O}\left(\frac{1}{N^2}\right) \right) \\ &= \langle q_{n+1} | p_n \rangle \exp \left[-\frac{i \Delta t}{\hbar} \left(\sum_i \frac{p_i^2}{2m} + V(q_i) \right) \right] \\ &= e^{i q_{n+1} p_n} \exp \left(-\frac{i \Delta t}{\hbar} H(q_{n+1}, p_n) \right) \end{aligned}$$

$$\begin{aligned} \prod_{n=0}^{N-1} \langle q_{n+1} | \hat{U}(\Delta t) | p_n \rangle &= \prod_{n=0}^{N-1} e^{i q_{n+1} p_n} e^{-\frac{i \Delta t}{\hbar} H(q_{n+1}, p_n)} \\ &= e^{i \sum_{n=1}^{N-1} q_{n+1} p_n} e^{-\frac{i \Delta t}{\hbar} \sum_{n=1}^{N-1} H(q_{n+1}, p_n)} + \mathcal{O}\left(\frac{1}{N}\right) \end{aligned}$$

$$\langle q_N | \hat{U}(t_N, t_i) | q_0 \rangle = \int \frac{d^D p_0}{(2\pi)^D} e^{-i p_0 q_0} \left[\int d\Omega e^{i \sum_{n=0}^{N-1} q_{n+1} p_n} e^{-\frac{i \Delta t}{\hbar} \sum_{n=0}^{N-1} H(q_{n+1}, p_n)} \times e^{-i \sum_{n=1}^{N-1} q_n p_n} \right]$$

$$= \int \frac{d^N p_0}{(2\pi\hbar)^N} \int d\Omega e^{\frac{i}{\hbar} \sum_{n=0}^{N-1} P_n (q_{n+1} - q_n)} e^{-\frac{i\Delta t}{\hbar} \sum_{n=0}^{N-1} H(q_{n+1}, p_n) + O\left(\frac{1}{N}\right)}$$

• $N \rightarrow \infty$, $\Delta t \rightarrow 0$, $\frac{1}{N}$ corrections vanish

$$\dot{q}_{n+1} \equiv \frac{q_{n+1} - q_n}{\Delta t}$$

$$\frac{i}{\hbar} \sum_{n=0}^{N-1} P_n (q_{n+1} - q_n) - \frac{i\Delta t}{\hbar} H(q_{n+1}, p_n)$$

$$\frac{i\Delta t}{\hbar} \sum_{n=0}^{N-1} \left(P_n \dot{q}_{n+1} - H(q_{n+1}, p_n) \right)$$

$$= \frac{i\Delta t}{\hbar} \sum_{n=0}^{N-1} L(q_{n+1}, p_n) \quad \left| \quad \begin{array}{l} \text{Define: } q(t_n) = q_n \\ p(t_n) = p_n \end{array} \right.$$

$$\sum_{n=0}^N \Delta t \xrightarrow[\Delta t \rightarrow 0]{N \rightarrow \infty} \int_{t_i}^{t_N} dt L(q(t), p(t))$$

$$S = \int_{t_0}^{t_N} dt L(q, p)$$

$$\langle q_f | \hat{U}(t_N, t_0) | q_i \rangle = \int \frac{d^N p_i}{(2\pi\hbar)^N} \int d\Omega e^{\frac{i}{\hbar} \int_{t_i}^{t_f} dt L(q(t), p(t))}$$

$$\int d\Omega = \lim_{N \rightarrow \infty} \prod_{n=1}^{N-1} \int \frac{d^N q_n d^N p_n}{(2\pi\hbar)^N}$$

$$\int DP(t) = \lim_{N \rightarrow \infty} \prod_{n=0}^{N-1} \int \frac{d^N p_n}{(2\pi\hbar)^N}$$

$$\int_{q_i}^{q_f} Dq(t) \equiv \lim_{N \rightarrow \infty} \prod_{n=1}^{N-1} \int d^N q_n$$

$$\frac{i}{\hbar} \int_{t_i}^{t_f} dt L(q(t), p(t))$$

$$\langle q_f | \hat{U}(t_f, t_i) | q_i \rangle = \int \mathcal{D}p(t) \int_{q_i} \mathcal{D}q e^{i \int_{t_i}^{t_f} p \dot{q} - H(p, q) dt}$$

(Phase-space path integral)